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DATA DESCRIPTOR

A digital phenotyping dataset for impending panic symptoms: a prospective longitudinal study

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This study investigated the utilization of digital phenotypes and machine learning algorithms to predict impending panic symptoms in patients with mood and anxiety disorders. A cohort of 43 patients was monitored over a two-year period, with data collected from smartphone applications and wearable devices. This research aimed to differentiate between the day before panic (DBP) and stable days without symptoms. With RandomForest, GradientBoost, and XGBoost classifiers, the study analyzed 3,969 data points, including 254 DBP events. The XGBoost model demonstrated performance with a ROC-AUC score of 0.905, while a simplified model using only the top 10 variables maintained an ROC-AUC of 0.903. Key predictors of panic events included evaluated Childhood Trauma Questionnaire scores, increased step counts, and higher anxiety levels. These findings indicate the potential of machine learning algorithms leveraging digital phenotypes to predict panic symptoms, thereby supporting the development of proactive and personalized digital therapies and providing insights into real-life indicators that may exacerbate panic symptoms in this population.

Background & Summary

Introduction. Panic symptoms are episodes of intense fear and anxiety that peak within minutes, along with various symptoms such as dyspnea, palpitations, dizziness, sweating, a sense of losing control, and a fear of death. Panic disorders are characterized by recurrent panic attacks and anticipatory anxiety regarding future panic symptoms¹. Patients with panic symptoms often face challenges as their intense physical and psychological symptoms are often mistaken for other non-panic somatic symptoms or medical issues, such as heart problems. This misinterpretation can lead to difficulties in daily functioning, quality of life, and work performance^{2,3}. Panic disorder has a prevalence of 2–5% and lifetime prevalence of 7–8%, whereas the prevalence of panic symptoms is relatively higher, ranging up to 14.4%^{4,5}. Furthermore, panic symptoms are not exclusive to panic disorder alone. Still, they can also occur as accompanying anxiety symptoms in mood and anxiety disorders^{6–8}.

Panic symptoms, often triggered by psychological distress or specific situations, lead individuals to avoid perceived causes of panic symptoms^{9,10}. However, similar to other psychiatric symptoms, it is important to understand panic symptoms in a multifactorial manner, such as neurophysiological, psychological, and environmental factors^{11,12}. Therefore, this comprehensive understanding enables integration of various causes and formulation of treatment strategies accordingly.

Previous studies reported that early intervention with psychoeducation, lifestyle changes, stress management, and relaxation training alleviate panic symptoms^{13,14}. In clinical settings, various approaches, including medications (e.g., antidepressants, anxiolytics) and non-pharmacological approaches (e.g., cognitive behavioral therapy, interoceptive exposures), are employed for treating panic symptoms. Despite considerable efforts, a substantial proportion of individuals still experience relapse after discontinuing medications, posing a challenge for continuous and consistent management of panic symptoms¹⁵. Current clinical setting, constrained by time and space issues, limits patients and medical experts from providing optimal medical attention. Although with regular doctor visits, evaluations during such appointments are fragmentary, limiting a thorough symptom assessment. Therefore, obtaining psychiatric digital phenotypic data, such as mood, anxiety, daily life activities,

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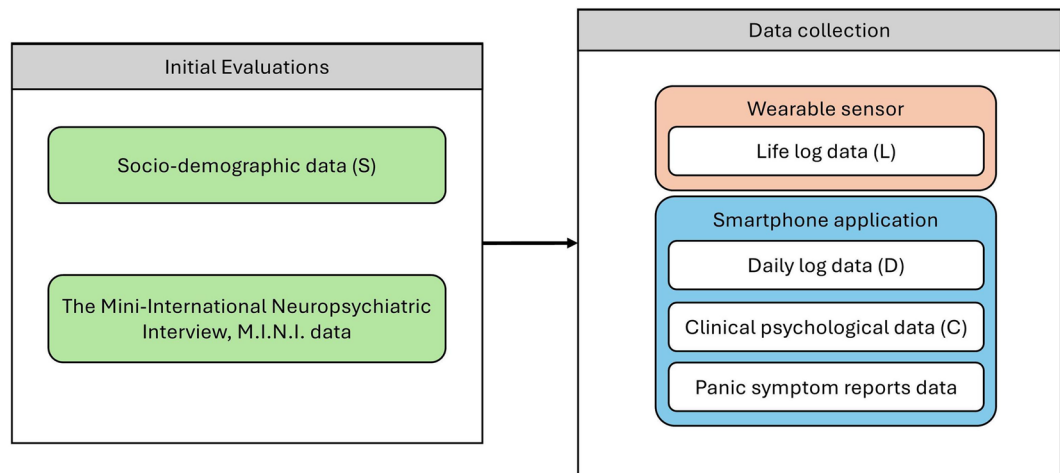


Fig. 1 Data collection protocol.

and sleep, can offer a multidimensional understanding, uncovering previously unnoticed differences among patients. Enabling individuals to visualize and monitor one's psychiatric state for panic symptoms will help patients alleviate anticipatory anxiety and empower proactive symptom management.

As digital devices like smartphones and wearable devices become universal, there's a growing interest in analyzing digital phenotypes – measurable data collected in real-time using a digital device in daily life – for psychiatric disorders¹⁶. These digital phenotypes, obtained through 24-hour continuous monitoring via digital devices, reflect an individual's state throughout the day and enable circadian rhythm analysis, offering the potential to predict the onset, aggravation, and relapse of psychiatric diseases^{17,18}. One study explored the potential of digital biomarkers by comparing wearable device data between days with and without panic attacks, finding that physiological parameters captured by wearable devices showed notable differences; nevertheless, the study did not proceed to develop predictive algorithms¹⁹. Similarly, a recent study used machine learning on wearable device data to predict the likelihood of panic symptoms occurring within a week; however, such long-term predictions could cause unnecessary anticipatory anxiety as it alerts patients a week before the expected onset of panic symptoms²⁰.

To address these gaps, we implemented a prospective digital phenotype cohort system, incorporating a smartphone application, inPHRsym (Softnet Co., Ltd., Seoul, Republic of Korea), while keeping track of panic symptoms and syncing with wearable devices in daily life. This study aimed to present a dataset of prospectively collected digital phenotypes to support the development of machine learning prediction models for predicting impending panic symptoms. By focusing on next-day predictions through 24-hour continuous monitoring from wearable devices, clinician-verified sociodemographic data, and a wide range of psychiatric indicators, we aim to minimize anticipatory anxiety while providing a more comprehensive and clinically relevant predictive model.

Methods

Ethical statement. All study procedures, approved by the Institutional Review Board of the Chungnam National University Sejong Hospital (Approval Number: 2020-12-016-007) and Korea University Anam Hospital (Approval Number: 2022AN0493), were conducted in accordance with the Declaration of Helsinki. Written consent was obtained from all patients after they were informed about the construction of the data collection platform and the sharing of their data. Consent was explicitly obtained through the inPHRsym smartphone application, where participants agreed to the data collection process and the sharing of de-identified data for research purposes before any data were collected.

Study design. This research was conducted as the Search Your Mind (S.Y.M., 심) project, using a smartphone application, inPHRsym, and wearable devices to collect personalized digital phenotypic data prospectively²¹. Data collection and participant inclusion followed the criteria outlined in our previously published protocol, adhering closely to the methods and procedures specified.

We designed our data collection strategy to integrate both state variables, which reflect short-term fluctuations in an individual's condition such as daily mood changes, and trait variables, which represent enduring characteristics like sex. We expect that this comprehensive approach allows us to capture the gradual accumulation of factors leading to panic symptoms, with state variables showing more pronounced changes as an individual approaches a panic event, while trait variables account for differences in baseline vulnerability across individuals, thus enabling us to estimate each patient's accumulating risk of panic onset.

As this is a prospective observational study, we aim to access the associations between multiple variables and the onset of panic symptoms. Each variable may show an association with either an increased or decreased risk of panic, depending on the context. These probabilistic associations accumulate across variables, contributing to an overall estimation of panic risk. By employing a multivariate approach that integrates data from various modalities, such as heart rate variability, anxiety levels, and mood fluctuations, we aim to accumulate these associations to enhance the prediction of panic event.

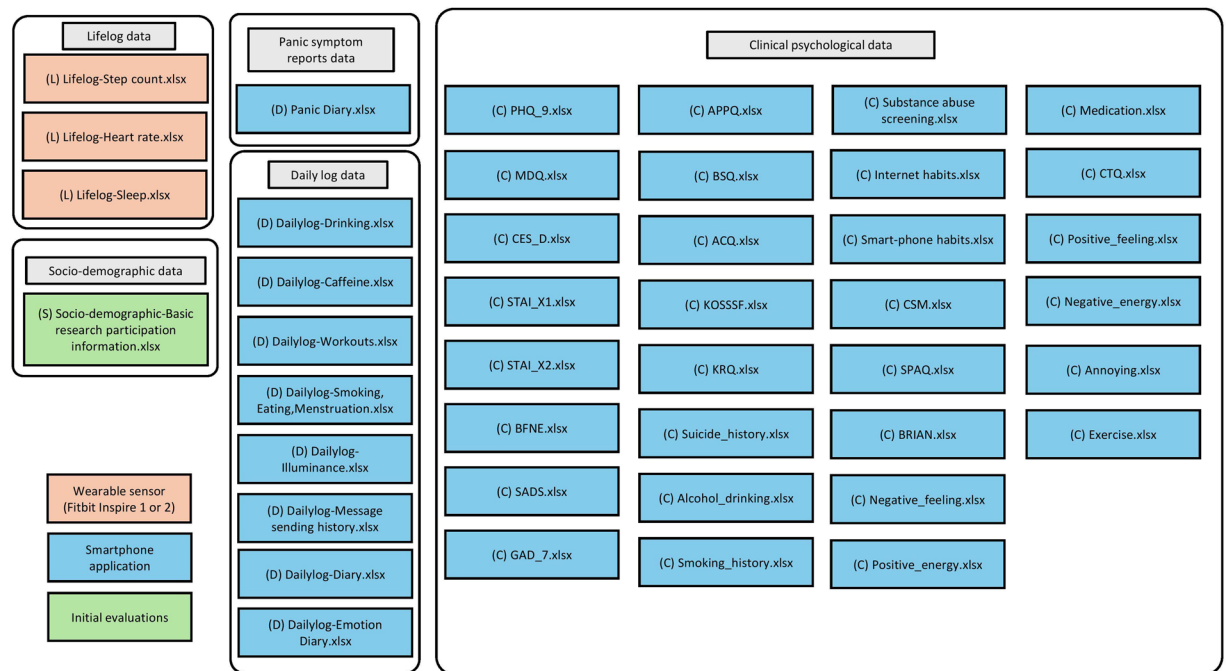


Fig. 2 Overview of inPHRsym dataset.

Participants. To obtain data centered on panic symptoms, the study population consisted of patients diagnosed with mood disorders (depressive disorder or bipolar disorder and related disorders), anxiety disorders, trauma and stressor-related disorders, and obsessive-compulsive and related disorders, according to the DSM-5 (Diagnostic and Statistical Manual of Mental Disorders) criteria. Those diagnosed with intellectual disabilities, organic brain disorders, neurocognitive disorders, or schizophrenic spectrum disorders were excluded.

Participants, aged 19–70, were outpatients in the Department of Psychiatry at Chungnam National University Sejong Hospital and Korea University Anam Hospital. They were recruited to participate after reporting mood or anxiety-related symptoms. Data from study participants were collected prospectively between May 2021 and Oct 2023.

The prospective digital phenotype-based cohort (S.Y.M., ∞ Project). The S.Y.M. project aimed to collect prospective digital phenotypic data centered on mood and anxiety symptoms in individuals with psychiatric disorders who met the inclusion criteria. Initial evaluations were conducted at the psychiatric clinic to collect socio-demographic data and to exclude individuals diagnosed with intellectual disabilities, organic brain disorders, neurocognitive disorders, or schizophrenic spectrum disorders. To confirm diagnoses and ensure participants met the necessary psychiatric criteria for inclusion, psychiatrist administered the Mini-International Neuropsychiatric Interview (M.I.N.I.), a structured diagnostic tool based on DSM and ICD-10 (International Classification of Diseases 10th Revision) criteria²². Subsequent to these initial evaluations, data collection was conducted using smartphone application connected to a wearable sensor (Fitbit Inspire 1 or 2) (Fig. 1).

Socio-demographic data. Socio-demographic items, including each participant's age, sex, height and weight, marital and job status, smoking, drinking, and suicide history, suicide attempts within one month, and psychiatric drug use within one month, were collected.

Digital phenotypic data. Life log data (heart rate, sleep, and step count data) were collected from the wearable devices and synchronized with the application. Fitbit devices, used as the wearable devices for collecting life log data, have been validated for accuracy and reliability in clinical research settings through multiple studies^{23–25}. Daily log data, which users reported each evening via the smartphone application, included mood status, energy levels, anxiety levels, annoyance levels, coffee consumption, alcohol consumption, smoking habits, and exercise status.

Panic symptom reports. inPHRsym requests users to report any panic symptoms experienced on a given day, including details such as the starting time, duration, symptom intensity, and accompanying symptoms such as palpitations and sweating, which are selected from a predefined list based on DSM-5 criteria (Supplementary Table 1). Since participants had already been diagnosed with panic disorder and their symptoms had been verified by psychiatric professionals as consistent with actual panic episodes, the self-reported panic reports were considered reliable.

Clinical psychological scales. Participants were instructed to respond periodically to multiple clinical psychological scales described in research protocols²¹. We analyzed the following clinical psychological scales: Patient Health Questionnaire-9, Generalized Anxiety Disorder-7, Center for Epidemiological Studies-Depression, Social

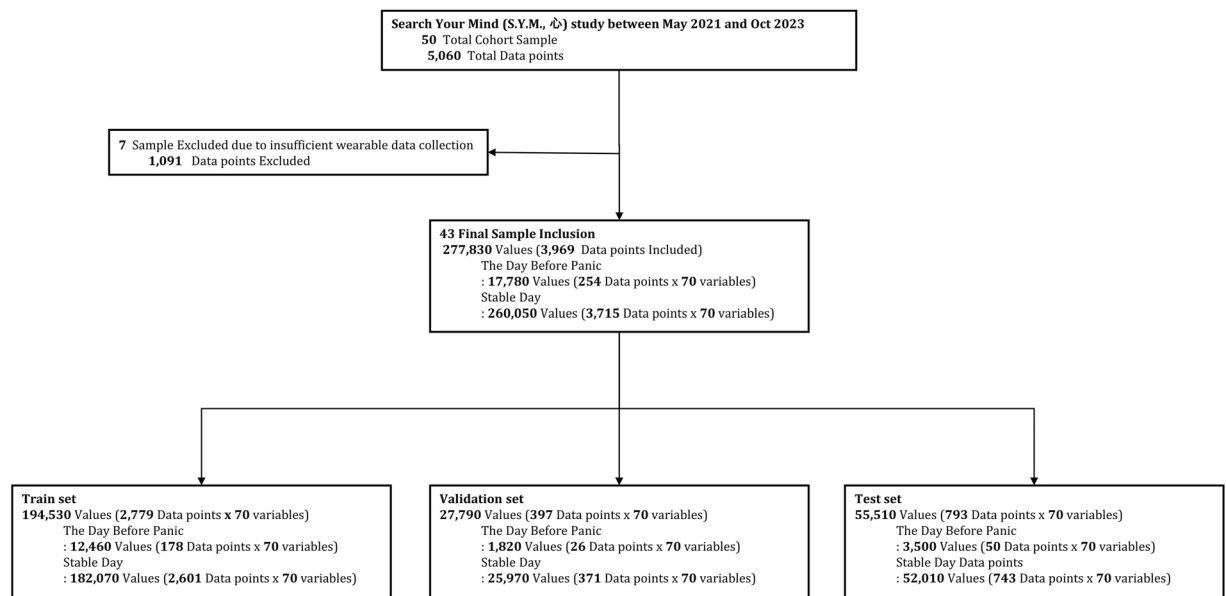


Fig. 3 Study design and workflow of participants selection.

Avoidance and Distress Scale, Anxiety Control Questionnaire, Albany Panic and Phobia Questionnaire, Korean Occupational Stress Scale-Short Form, Brief Fear of Negative Evaluation, Body Shape Questionnaire, Composite Scale for Morningness, Biological Rhythms Interview for Assessment in Neuropsychiatry, Mood Disorder Questionnaire, State-Trait Anxiety Inventory, Korean Resilience Quotient Test, Seasonal Pattern Assessment Questionnaire, and Childhood Trauma Questionnaire^{26–41}.

Data Records

The impending panic symptom dataset is openly available on Zenodo under a CC BY license⁴². The data consists of 44 files that can be categorized into four clusters (Lifelog data, Socio-demographic data, Panic symptom reports data, Daily log data, Clinical psychological data) (Fig. 2). The dataset includes socio-demographic data collected through direct clinical assessments at the hospital, which is organized in file 'Basic research participation information.xlsx'. Socio-demographic data collected through clinical assessment by psychiatric professional is stored in the file named 'Socio-demographic-Basic research participation information.xlsx'. Life log data, including heart rate, sleep, and step count, were gathered from wearable device and stored in files named 'Lifelog-Step count.xlsx', 'Lifelog-Heart rate.xlsx', and 'Lifelog-Sleep.xlsx'. Additionally, users entered daily log data every evening using the smartphone application, which included details about their mood, energy levels, anxiety states, annoyance states, coffee intake, alcohol use, smoking habits, and exercise activities. These reports are documented in files named 'Dailylog-Drinking.xlsx', 'Dailylog-Caffeine.xlsx', 'Dailylog-Workouts.xlsx', 'Dailylog-Smoking, Eating, Menstruation.xlsx', 'Dailylog-Illuminance.xlsx', 'Dailylog-Message sending history.xlsx', 'Dailylog-Diary.xlsx', and 'Dailylog-Emotion Diary.xlsx'. Panic symptoms experienced on a given day, including specific symptoms such as palpitations and sweating, were reported by users through the smartphone application and documented in the file named 'Panic Diary.xlsx'. Clinical psychological data, collected through regular participant responses to multiple scales as outlined in the research protocols, are stored in various files named 'PHQ_9.xlsx', 'MDQ.xlsx', 'CES_D.xlsx', 'STAI_X1.xlsx', 'STAI_X2.xlsx', 'BFNE.xlsx', 'SADS.xlsx', 'GAD_7.xlsx', 'APPQ.xlsx', 'BSQ.xlsx', 'ACQ.xlsx', 'KOSSSE.xlsx', 'KRQ.xlsx', 'Suicide_history.xlsx', 'Alcohol_drinking_history.xlsx', 'Smoking_history.xlsx', 'Substance abuse screening.xlsx', 'Internet habits.xlsx', 'Smart-phone habits.xlsx', 'CSM.xlsx', 'SPAQ.xlsx', 'BRIAN.xlsx', 'Negative_feeling.xlsx', 'Positive_energy.xlsx', 'Medication.xlsx', 'CTQ.xlsx', 'Positive_feeling.xlsx', 'Negative_energy.xlsx', 'Annoying.xlsx', and 'Exercise.xlsx'. Detailed description of each group is available at Supplementary Table 9.

Data preprocessing. In this study, since the data of the day became a single data point, the data were statistically processed to express the status of each day. Data analysis was limited to cases where heart rate data was available for more than 12 h within a single day. For each day, the mean, maximum, and variance of heart rate and step count were computed. Additionally, hourly variances were averaged across the day to generate variables representing heart rate and step count variability (e.g., HR_hvar_mean, steps_hvar_mean).

As there are previous reports that panic symptoms show daily variation and seasonality, we applied Cosinor analysis, which fits time-series data to a cosine function, to examine circadian rhythms in our study⁴³. Lifelog data collected from wearable devices, such as heart rate, activity, and sleep data, were processed using this method to derive their daily acrophase, amplitude, and mesor values, demonstrating the role of these physiological parameters in influencing panic symptoms⁴⁴. To capture shifts in circadian rhythm, differences in these values were computed relative to the previous day and two days prior (e.g., HR_acrophase_difference, HR_acrophase_difference_2d).

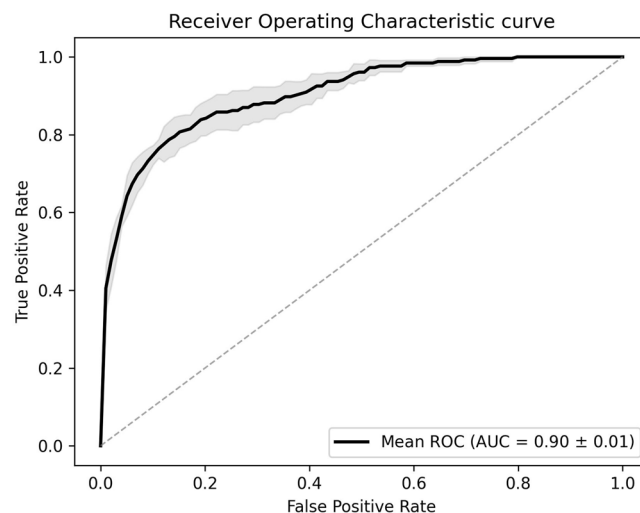


Fig. 4 Receiver operating characteristic curve (ROC) of panic prediction model using XGBoost classifier. The confidence band shows one standard deviation above and below the mean ROC curve of 5-fold cross-validation. Abbreviation: ROC, Receiver operating characteristic curve; AUC, Area under the receiver operating characteristic curve; CI, Confidence Interval.

In addition, band power analysis was conducted to quantify the heart rate change pattern to process the four band power variables calculated by the wavelength bands. A total of 1440 heart rate data points per day were transformed via Fourier transformation, and four distinct band power values were computed using multi-taper methods across specified wavelength bands: 0.001–0.0005 Hz, 0.0005–0.0001 Hz, 0.0001–0.00005 Hz, 0.00005–0.00001 Hz. As a result, 70 items were selected as variables (Supplementary Table 3).

Missing values encountered during the calculation of averages and variances were excluded from the analysis to prevent bias. In the context of Cosinor and band power analyses, where missing data would inhibit calculation, interpolation methods were employed to estimate and replace missing values. To ensure comparability across different variables and enhance model performance, all continuous data were subjected to normalization and standardization before being utilized in the predictive models.

Model construction and evaluation. The data points on the day of a panic event were labeled as class “the day of panic (DP)” and on the day before panic event were labeled as class “the day before panic (DBP)”. Data points of the day without a panic event the following day were labeled class “stable day (SD)”. The classification performance of our model was evaluated using 5-fold cross-validation. For each cross-validated fold, the data were randomly divided into a training set, a validation set, and a testing set at a ratio of 7:1:2. These data were then learned using RandomForest(version 1.3.2), GradientBoost(version 1.3.2), and XGBoost(version 1.7.4) classifiers to distinguish DBP and SD. In terms of clinical applicability, if DBP can be predicted based on digital phenotypes, patients experiencing panic symptoms will be able to prepare for and manage their panic the following day more proactively. Furthermore, by identifying an “SD” when no panic symptoms are expected to occur the following day, we can reassure patients about their reduced likelihood of experiencing panic symptoms, thereby alleviating their anticipatory anxiety. Therefore, we aimed to predict the DBP and SD of patients.

To evaluate the model performance, the AUROC, accuracy, prediction, and recall scores were calculated. In addition, Shapley values (version 0.44.0), which estimate the contribution of the variables in the machine-learning model, were calculated. Shapley values are computed for each sample, and these sample-specific results were independently plotted, resulting in beeswarm plots. To represent the importance of each variable, the absolute values of the sample-specific results were averaged for each variable and displayed as a bar plot. Additionally, to address the issue of data imbalance in our dataset, we employed various sampling techniques, such as SMOTE (Synthetic Minority Oversampling Technique), undersampling, and adjusted sample weights during model training and compare the result with each technique (Supplementary Table 4, 5).

Data ablation. We trained models separately on clinical, daily log, and lifelog data, constructing three distinct models. The performance scores of these models were evaluated, and Shapley values were derived from each model to identify influential variables within the respective data types. To explore the potential for a lightweight model, we also constructed a dataset of the top 10 highest Shapley values and evaluated the performance of the model trained on this limited dataset.

Technical validation

Clinical characteristics of participants. Between May 2021 and Oct 2023, 5,060 data points were collected from 50 patients with mood and anxiety symptoms at the psychiatric clinics of Chungnam National University Sejong Hospital and Korea University Anam Hospital. After excluding patients with insufficient wearable data collection, a total of 43 participants, consisting of 18 men and 25 women, were included in the analysis.

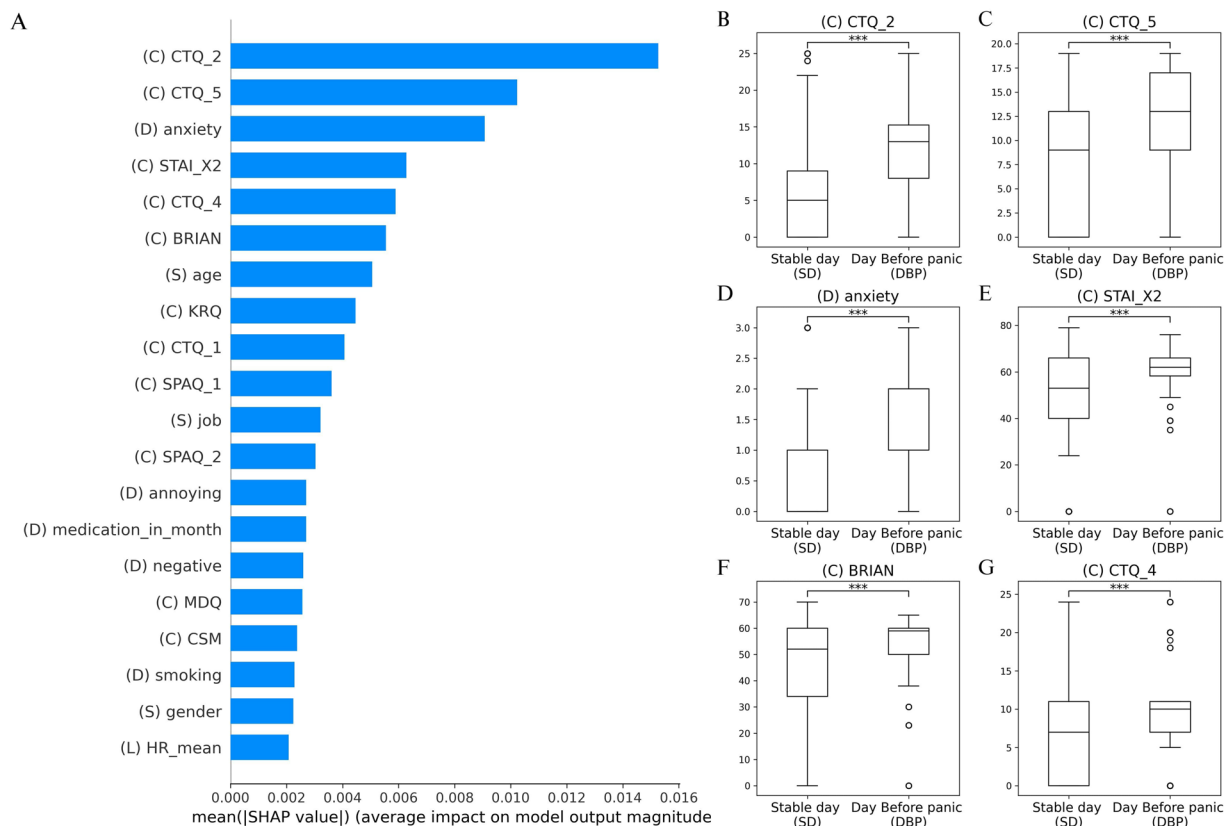


Fig. 5 (A) The top 20 variables with the highest Shapley values from the XGBoost classifier trained on the entire digital phenotypic dataset and (B–G) box plots displaying the distribution of the 6 variables with the highest Shapley values for DBP and SD. Abbreviations: ‘(C)’, Clinical psychological data; ‘(D)’, Daily log data; ‘(S)’, Socio-demographic data; ‘(L)’, Life log data; CTQ, Childhood Trauma Questionnaire; STAI-X2, State-Trait Anxiety Inventory score – Trait Anxiety section; BRIAN, Biological Rhythms Interview of Assessment in Neuropsychiatry; KRQ-53, Korean Resilience Quotient-53; SPAQ, Seasonal Pattern Assessment Questionnaire; MDQ, Mood Disorder Questionnaire; CSM, Composite Scale of Morningness; SD, the stable day; DBP, the day before panic; ns, statistically no significant; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

The final dataset comprised 3,969 data points from these 43 patients, with 70 variables and 277,830 values (Fig. 3). The average age was 36.0 years ($SD = 11.3$). 25 participants out of 43 people (55.6%) were prescribed with psychiatric medications. Among the analysis subjects, 23 (54.8%) reported being diagnosed with depressive disorder, 11 (26.2%) reported being diagnosed with bipolar disorder, 15 (35.7%) reported being diagnosed with panic disorder, 6 (14.3%) reported being diagnosed with society anxiety disorder. (Supplementary Table 6) The demographic and clinical characteristics of our cohort, including the high comorbidity of mood disorders, the age prevalence, and the larger proportion of women align with previously reported patterns in panic disorder populations, supporting the applicability of our dataset to the broader population of individuals with panic symptoms^{8,45–47}. A total of 3,969 data points was collected, with 254 corresponding to the day before panic events (DBP) and the remaining 3,715 labeled as stable days (SD).

Performance of panic event prediction model. The ROC-AUC scores of the GradientBoost Classifier, RandomForest Classifier, and XGBoost Classifier were found to be 0.826 (95% CI 0.736–0.917), 0.730 (95% CI 0.649–0.811), and 0.905 (95% CI 0.885–0.924), respectively (Supplementary Figure 1, Fig. 4). Among the three models, the XGBoost Classifier had the highest ROC-AUC score of 0.905, and its accuracy score was 0.919 (95% CI 0.905–0.934). (Fig. 4, Supplementary Table 7) These results indicate that the XGBoost model can effectively distinguish between data points from the DBP and SD, demonstrating that our digital phenotyping dataset contains valuable information for identifying impending panic events. The model’s ability to predict these impending panic events could potentially contribute to reducing anticipatory anxiety in individuals prone to panic symptoms. We also compare the result using various sampling techniques to our model, and we found that XGBoost’s performance using class weight is superior overall (Supplementary Table 4).

Explainability of the AI model. To identify the most influential variables in the XGBoost panic prediction model, the mean absolute Shapley values were calculated for each variable. The top 20 variables with the highest Shapley values were visualized (Fig. 5A), revealing that variables from the clinical data and daily log data had greater impact on the model’s predictions. In particular, CTQ, anxiety level, BRIAN, and STAI had high Shapley

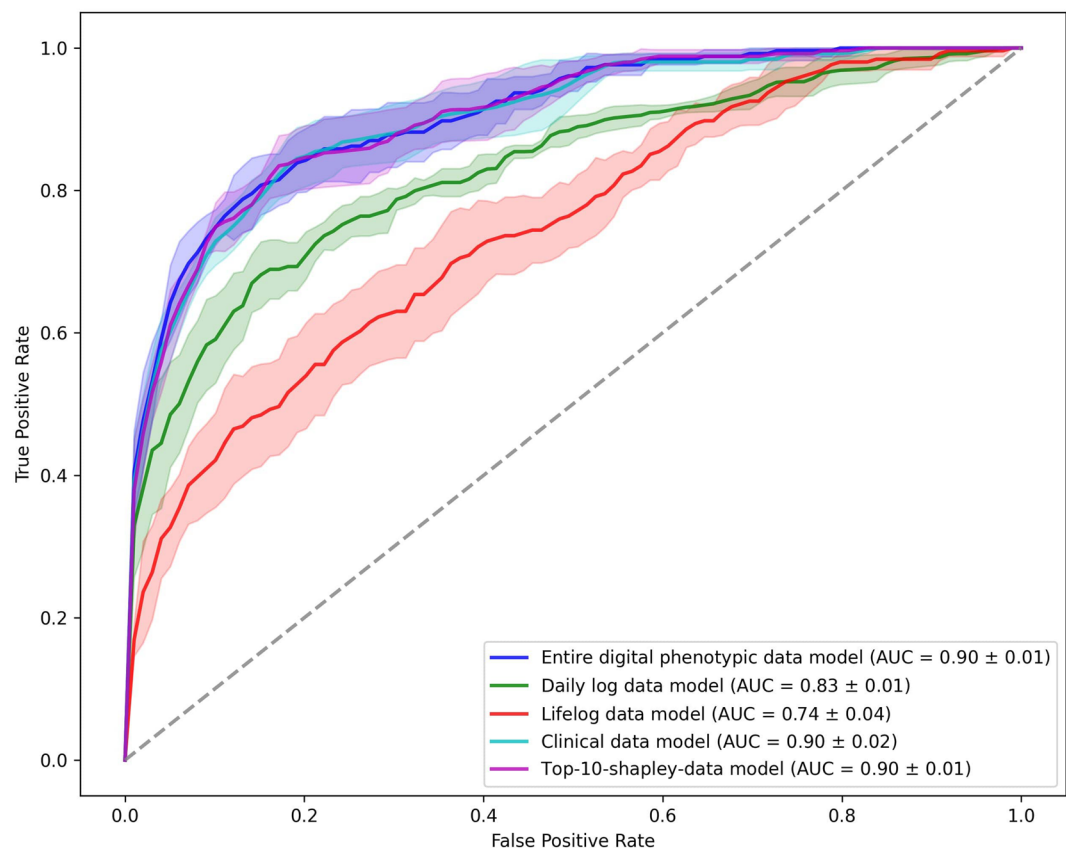


Fig. 6 Receiver operating characteristic curve (ROC) of entire digital phenotypic data model, daily log data model, life log data model, clinical data model, and Top-10-Shapley-data model. The confidence band shows one standard deviation above and below the mean ROC curve of 5-fold cross-validation. Abbreviation: ROC, Receiver operating characteristic curve; AUC, Area under the receiver operating characteristic curve; CI, Confidence Interval.

values. The actual values of these variables were found to be significantly higher (p -value < 0.001) on the DBP compared to SD (Fig. 5B–G).

Performance of model with data ablation. To assess the predictive performance of different data types, we built separate models using only clinical, daily log, life log data. Additionally, we constructed a dataset consisting of the top 10 variables with the highest Shapley values from the entire digital phenotypic data model. The clinical data model, which utilized participants' self-reported responses to multiple clinical psychological scales, demonstrated the highest ROC-AUC score of 0.900 (95% CI 0.876–0.925) among the three models. The daily log data model, which included various self-reported measures such as anxiety levels and energy levels, achieved a performance of 0.827 (95% CI 0.816–0.839). Notably, using only life log data, composed of heart rate, sleep, and step count data, the model could predict the occurrence of panic symptoms the following day with an ROC-AUC score of 0.745 (95% CI 0.693–0.796) (Fig. 6, Supplementary Table 8).

The Top-10-Shapley-data model achieved an ROC-AUC score of 0.903 (95% CI 0.883–0.922), which was comparable to the entire digital phenotypic data model (ROC-AUC 0.905, 95% CI 0.885–0.924) (Fig. 6). However, the accuracy score of the Top-10-Shapley-data model was 0.838 (95% CI 0.824–0.853), which was lower than that of the entire digital phenotypic data model (Supplementary Table 8). These results suggest the possibility of a lightweight model that offers fast and efficient prediction by using selected variables when all comprehensive data cannot be collected in the real world.

Explainability of model with data ablation. We examined the importance of variables on panic prediction in the entire digital phenotypic data model, daily log data model, life log data model, a clinical data model using Shapley values visualized as beeswarm plots (Fig. 7). In a beeswarm plot, each dot represents a sample's Shapley value for a specific variable, and a wider distribution of points indicates a higher importance of corresponding variable in the model. In each model, 'CTQ_2', 'anxiety', and 'steps_mean' showed the widest distribution, indicating they were the most important variables in each model. In these beeswarm plots, if a variable contributes to the prediction of a panic event (DBP), it will have a positive value, while a variable contributing to the prediction of a stable day (SD) will have a negative value. 'CTQ_2', 'anxiety', and 'steps_mean' have red dots with positive values, suggesting that increased step counts, elevated anxiety levels, and higher Childhood Trauma Questionnaire scores are associated with the occurrence of panic events the following day. In contrast,

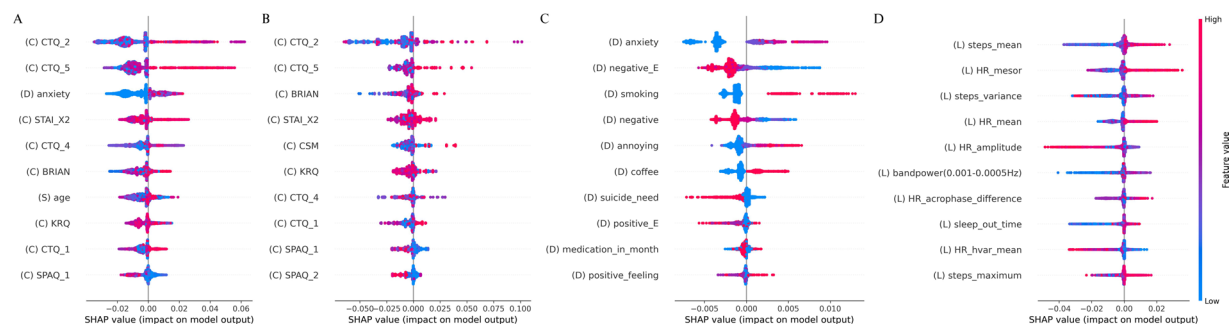


Fig. 7 Beeswarm plots of sample-specific Shapley values in the entire digital phenotypic data model (A), clinical data model (B), daily log data model (C), and life log data model (D). In these plots, each point represents the Shapley value of a particular variable for an individual sample. A wider distribution of points indicates the higher importance of corresponding variable in the model. Positive Shapley values, positioned on the right side of the plot, indicate variables that contribute to the prediction of a DBP, whereas negative Shapley values, located on the left side, correspond to variables that are predictive of SD. Abbreviations: ‘(C)’; Clinical psychological data; ‘(D)’; Daily log data; ‘(S)’; Socio-demographic data; ‘(L)’; Life log data; CTQ, Childhood Trauma Questionnaire; STAI-X2, State-Trait Anxiety Inventory score – Trait Anxiety section; BRIAN, Biological Rhythms Interview of Assessment in Neuropsychiatry; KRQ-53, Korean Resilience Quotient-53; SPAQ, Seasonal Pattern Assessment Questionnaire; MDQ, Mood Disorder Questionnaire; CSM, Composite Scale of Morningness; SD, the stable day; DBP, the day before panic.

‘negative_E’ (negative energy, ranging from 0 to -4 , with more severe negative energy indicated by lower values) has blue dots with positive values, indicating that more severe negative energy is associated with the occurrence of panic events.

Usage Notes

This study presents a unique dataset of digital phenotypic data collected prospectively from total in 5,060 days collected prospectively in real-time from 43 participants over two years. The dataset was prospectively collected from the Search Your Mind (S.Y.M., <https://www.searchyourmind.com/>) project with the inPHRsym smartphone application, using a wearable device²¹. We aimed to overcome the limitations of existing psychiatric research by allowing the real-time recording of everything from lifelogs to daily logs, psychological scales, and panic symptoms, as recall bias is a frequent problem in psychiatric research. Socio-demographic data was collected through psychiatric clinic visits, ensuring accurate baseline information from medical professionals. Wearable device continuously gathered comprehensive lifelog data, providing objective physiological measures throughout the day. A smartphone application was used to collect daily logs, clinical psychological scales and panic symptoms reports, maximizing the capture of patients’ individual subjective symptom experiences.

We believe that our digital phenotypic dataset that contributes to the prediction of panic symptoms may provide a meaningful step not only towards developing algorithms to minimize patient’s anticipatory anxiety but also towards a deeper understanding of panic itself. This approach may lead to a more detailed understanding of panic symptoms and their triggers, enabling proactive personalized interventions and digital therapeutics. Utilizing digital phenotypic dataset to understand the essence of panic symptoms will help both patients and mental health professionals better assess and manage individuals suffering from panic symptoms.

Several limitations of this dataset should be acknowledged for the potential users. Some measures, such as daily mood and energy levels, depend on self-reported data, which can lead to potential bias in symptom reporting and its severity assessment. Differences exist in the number of data points per participant exist, which is an inherent characteristic of real-world data collection in prospective observational studies. This variability reflects the naturalistic setting of the study but may potentially introduce model bias and influence the reliability of individual-level analyses. Despite these limitations, this study offers a rich dataset with the potential to develop more effective and personalized interventions for individuals suffering from panic symptoms. The result could provide evidence to support proactive and personalized digital therapeutics, offering a comprehensive view of panic symptoms through real-time, multimodal data collection in real-world settings.

Code availability

Project home page: <https://github.com/DigitalHealthcareLab/24PanicPrediction>.

Programming language: Python 3

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Competing interests

The authors declare no competing interests.

Additional information

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